

Lost & Found System

EC2 1

1. API Gateway

2. Auth

POST	/auth/register	Belong to Registration micro
POST	/auth/login	Belong to Authentication micro

SCHEMA:

```
{
  id: ObjectId,
  name: String,
  email: String,
  password: String,
  role: String,      (student or admin)
  createdAt: Date
}
```

3. Catalog Service

GET	/items	Get all items in the system
GET	/items/:id	Get item by id
GET	/items/lost	Get all lost items
GET	/items/found	Get all found items
POST	/items/	Posting a lost item (student/admin)
PUT	/admin/items/:id	Update info of item (admin only)
DELETE	/admin/items/:id	Delete item, <i>can be soft delete?</i> (admin only)

SCHEMA:

```
{
  id: ObjectId,
  title: String,
  description: String,
  category: String,
  status: String,      (lost or found)
  location: String,
  reportedBy: ObjectId, (from user id)
}
```

```
    createdAt: Date
  }
}
```

4. Claims Service

GET	/admin/claims	Get all claim
GET	/claims/:id	Get a claim by id
GET	/student/my-claims	Student get their own claims
POST	/student/submit-claim	Submit a claim (student)
PUT	/admin/approve-claim/:id	Approve a claim from student (admin)
PUT	/admin/reject-claim/:id	Reject a claim from student (admin)

SCHEMA:

```
{
  id: ObjectId,
  itemId: ObjectId,
  claimedBy: ObjectId,
  status: String,
  message: String,
  reviewedBy: ObjectId,
  createdAt: Date,
  updatedAt: Date
}
```

5. Report Service

GET	/admin/total-items	Get all lost & found items
GET	/admin/total-claims	Get all claim submissions
GET	/admin/pending-claims	Get all pending claims
GET	/admin/claims-by-status	Approved / Rejected / Pending

6. Activity Log service

POST	/log-action (this is internal)	Log an action (called by other services automatically)
GET	/admin/all-log	Admin sees all activity logs
GET	/admin/logs/:userId	Admin sees logs by specific user

SCHEMA:

```
{
  "id": "ObjectId",
  "userId": "ObjectId",
  "action": "String",
  "description": "String",
  "createdAt": "Date"
}
```

7. Feedback Service

POST	/student/submit-feedback	Submit feedback
GET	/admin/feedback	Get all feedback (admin)

SCHEMA:

```
{
  "id": "ObjectId",
  "userId": "ObjectId",
  "message": "String",
  "createdAt": "Date"
}
```

All services are comprised of either docker container(s) (if multiple containers, must be used with compose).